

# A sharp radial one-shot bound via Gamma–Beta submultiplicativity

Candidate solution packet for arXiv:2405.07253

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**Status.** Candidate full solution, likely valid, pending human review. The result completely answers the motivating radial one-shot question on page 23 of the source paper. It does *not* prove the stronger consecutive-monotonicity statement called Conjecture 2 there.

## 1 Source question

The source is S. Brazitikos and G. Chasapis, *Sharp estimates for the Cramér transform of log-concave measures and geometric applications*, arXiv:2405.07253, published in *Advances in Mathematics* 478 (2025), 110407 [1]. On PDF page 23, immediately after Corollary 4.18, the authors ask whether

$$\mathbb{E}e^{-\Lambda_X^*(X)} \leq (\mathbb{E}e^{-\Lambda_Y^*(Y)})^n \quad (1)$$

holds for every rotationally invariant, not necessarily product, log-concave random vector  $X$  in  $\mathbb{R}^n$ , where  $Y$  is the standard one-dimensional double exponential. They then propose the stronger Conjecture 2: if  $X_n$  is the isotropic radial exponential in dimension  $n$ , then  $n \mapsto (\mathbb{E}e^{-\Lambda_{X_n}^*(X_n)})^{1/n}$  is non-increasing.

## 2 Exact one-dimensional reduction

For  $n \geq 1$ , let  $\mathcal{E}_n$  denote the isotropic radial exponential law on  $\mathbb{R}^n$ , whose density is proportional to  $\exp(-\sqrt{n+1}|x|)$ , and set

$$A_n := \mathbb{E} \exp(-\Lambda_{\mathcal{E}_n}^*(\mathcal{E}_n)).$$

The explicit Cramér transform computed in the source paper gives

$$\Lambda_{(\mathcal{E}_n)_1}^*(r) = \frac{n+1}{2} h\left(\frac{r}{\sqrt{n+1}}\right), \quad h(q) := \sqrt{1+4q^2} - 1 - \log\left(\frac{\sqrt{1+4q^2} + 1}{2}\right). \quad (2)$$

If  $R_n = |\mathcal{E}_n|$  and  $S_n = \sqrt{n+1} R_n$ , polar integration shows that  $S_n$  has the Gamma law  $\text{Gamma}(n, 1)$ . Consequently

$$A_n = \mathbb{E}e^{-F_n(S_n)}, \quad F_n(s) := \frac{n+1}{2} h\left(\frac{s}{n+1}\right). \quad (3)$$

## 3 Proof intuition

The Gamma representation (3) is compatible with addition of dimensions: independent Gamma variables of shapes  $n$  and  $m$  add to a Gamma variable of shape  $n+m$ . Conditional on their sum,

from positive to negative, note that  $\phi(x) = \frac{\phi(\varphi(x)) - \phi(x)}{\varphi(x) - x}$ , where  $\varphi(x) = (\Lambda_{\mathcal{E}_1}^*)'(\varphi(x))$ . Since  $\phi'(x) = \frac{(\Lambda_{\mathcal{E}_1}^*)''(x)x - (\Lambda_{\mathcal{E}_1}^*)'(x)}{x^2} \leq 0$  (again, by Corollary 4.8) we get that  $\phi$  is non-increasing, with  $\phi(x) \rightarrow 1$  as  $x \rightarrow 0$ . Since  $a < 1/2$ , it follows then that  $\phi(x) > 2a$  holds only for  $x$  sufficiently close to 0, which shows the desired result for  $\Phi'$ .  $\square$

As a direct corollary of Theorem 4.13, for the case  $n = 1$ , we get the following bound on the exponential one-shot separability constant of *symmetric* log-concave distributions.

**Corollary 4.18.** *Let  $X = (X_1, \dots, X_n)$  be a random vector in  $\mathbb{R}^n$  with independent and symmetric log-concave components. Then  $\mathbb{E}e^{-\Lambda_X^*(X)} \leq (\mathbb{E}e^{-\Lambda_Y^*(Y)})^n$ , where  $Y$  is a standard double exponential (note that  $\mathbb{E}e^{-\Lambda_Y^*(Y)} \simeq 0.787\dots$ ).*

*Proof.* The case  $n = 1$  of Theorem 4.13 implies that  $\mathbb{E}e^{-\Lambda_{X_j}^*(X_j)} \leq \mathbb{E}e^{-\Lambda_Y^*(Y)}$  for every  $j = 1, \dots, n$ , and by independence we have that  $\mathbb{E}e^{-\Lambda_X^*(X)} = \prod_{j=1}^n \mathbb{E}e^{-\Lambda_{X_j}^*(X_j)}$ .  $\square$

Given Theorem 4.13, one could wonder if a more general result than that of Corollary 4.18 holds, namely that the right-hand side bound  $(\mathbb{E}e^{-\Lambda_Y^*(Y)})^n$  is true for all rotationally-invariant (not necessarily product) log-concave distributions  $X$  in  $\mathbb{R}^n$ . We already know that for every such  $X$ ,  $\mathbb{E}e^{-\Lambda_X^*} \leq c^n$  for some absolute constant  $c \in (0, 1)$  (this follows already from (12) and the well known fact that  $L_X$  is bounded in the rotationally invariant case). We believe that the sharp value of  $c$  is that provided by Theorem 4.13. In particular, numerical evidence leads us to the following conjecture. To stress the dependence on the dimension  $n$ , we denote here by  $X_n$  the random vector  $\mathcal{E}$  in  $\mathbb{R}^n$ .

*Conjecture 2.* The function

$$n \mapsto \left( \mathbb{E}e^{-\Lambda_{X_n}^*(X_n)} \right)^{\frac{1}{n}}$$

is non-increasing.

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Figure 1: The question and Conjecture 2 on page 23 of the source PDF.

the splitting proportion is  $\text{Beta}(n, m)$ . Thus the product  $A_n A_m$  becomes a conditional average over one Beta variable.

There are two Jensen steps. First,  $x \mapsto e^{-x}$  is convex, so averaging over the random Beta split can only increase the exponential relative to the exponential of the averaged energy. Second, the derivative  $h'$  is concave. After size-biasing the Beta split, its mean lands at exactly the scale  $1/(n + m + 1)$  occurring in  $F_{n+m}$ . This proves that the averaged split energy is at most the unsplit energy  $F_{n+m}$ . The two steps yield  $A_{n+m} \leq A_n A_m$ .

## 4 The submultiplicativity theorem

**Lemma 1** (Concavity of the derivative). *The function  $h'$  is strictly concave on  $[0, \infty)$ .*

*Proof.* Writing  $w = \sqrt{1 + 4q^2}$ , direct differentiation of (2) gives

$$h'(q) = \frac{4q}{w+1}, \quad h''(q) = \frac{4}{w(w+1)},$$

and hence

$$h'''(q) = -\frac{16q(2w+1)}{w^3(w+1)^2} < 0 \quad (q > 0).$$

Thus  $h'$  is concave on  $[0, \infty)$  and strictly concave away from the endpoint.  $\square$

**Lemma 2** (Beta splitting inequality). *Let  $n, m \geq 1$  and let  $D \sim \text{Beta}(n, m)$ . Then, for every  $s \geq 0$ ,*

$$\mathbb{E}[F_n(sD) + F_m(s(1-D))] \leq F_{n+m}(s). \quad (4)$$

*The inequality is strict when  $s > 0$ .*

*Proof.* Let the right-hand side of (4) minus the left-hand side be  $G(s)$ . Clearly  $G(0) = 0$ , and (3) gives  $F'_k(s) = \frac{1}{2}h'(s/(k+1))$ . Therefore

$$\begin{aligned} 2G'(s) &= h'\left(\frac{s}{n+m+1}\right) \\ &\quad - \mathbb{E}\left[Dh'\left(\frac{sD}{n+1}\right) + (1-D)h'\left(\frac{s(1-D)}{m+1}\right)\right]. \end{aligned} \quad (5)$$

The elementary size-bias identities for a Beta variable say

$$\begin{aligned} \mathbb{E}[D\varphi(D)] &= \frac{n}{n+m}\mathbb{E}\varphi(D_+), & D_+ &\sim \text{Beta}(n+1, m), \\ \mathbb{E}[(1-D)\varphi(1-D)] &= \frac{m}{n+m}\mathbb{E}\varphi(E_+), & E_+ &\sim \text{Beta}(m+1, n). \end{aligned}$$

Their means are

$$\mathbb{E}D_+ = \frac{n+1}{n+m+1}, \quad \mathbb{E}E_+ = \frac{m+1}{n+m+1}.$$

By Lemma 1 and Jensen's inequality,

$$\begin{aligned} \mathbb{E}h'\left(\frac{sD_+}{n+1}\right) &\leq h'\left(\frac{s}{n+m+1}\right), \\ \mathbb{E}h'\left(\frac{sE_+}{m+1}\right) &\leq h'\left(\frac{s}{n+m+1}\right). \end{aligned}$$

After multiplication by  $n/(n+m)$  and  $m/(n+m)$ , these two estimates show from (5) that  $G'(s) \geq 0$ . Since  $G(0) = 0$ , (4) follows. Strict concavity and the non-degeneracy of the Beta laws give strictness for  $s > 0$ .  $\square$

**Theorem 1** (Submultiplicativity). *For all integers  $n, m \geq 1$ ,*

$$A_{n+m} \leq A_n A_m. \quad (6)$$

*In fact, the inequality is strict.*

*Proof.* Let  $U \sim \text{Gamma}(n, 1)$  and  $V \sim \text{Gamma}(m, 1)$  be independent. The standard Gamma–Beta change of variables gives

$$S := U + V \sim \text{Gamma}(n+m, 1), \quad D := \frac{U}{U+V} \sim \text{Beta}(n, m),$$

and  $S, D$  are independent. (This also follows directly by applying the Jacobian  $s$  to  $u = sd$ ,  $v = s(1 - d)$ .) By (3), conditional Jensen for the convex function  $x \mapsto e^{-x}$ , and Lemma 2,

$$\begin{aligned} A_n A_m &= \mathbb{E} e^{-F_n(U) - F_m(V)} \\ &= \mathbb{E}_S \mathbb{E}_D e^{-F_n(SD) - F_m(S(1-D))} \\ &\geq \mathbb{E}_S \exp(-\mathbb{E}_D [F_n(SD) + F_m(S(1-D))]) \\ &\geq \mathbb{E}_S e^{-F_{n+m}(S)} = A_{n+m}. \end{aligned}$$

Strictness holds because  $S > 0$  almost surely and the strict part of Lemma 2 applies.  $\square$

## 5 Full answer to the radial one-shot question

**Corollary 1** (Sharp all-dimensional radial constant). *Let  $X$  be a full-dimensional rotationally invariant log-concave random vector in  $\mathbb{R}^n$ , and let  $Y$  be the isotropic one-dimensional double exponential. Then*

$$\mathbb{E} e^{-\Lambda_X^*(X)} \leq A_n \leq A_1^n = (\mathbb{E} e^{-\Lambda_Y^*(Y)})^n. \quad (7)$$

*Thus the question preceding Conjecture 2 in the source paper has an affirmative answer. The base  $A_1 \simeq 0.7872717101$  is sharp for a bound required to hold in every dimension  $n \geq 1$ .*

*Proof.* The first inequality in (7) is Theorem 4.13 of the source paper [1], whose extremizer among radial log-concave laws in dimension  $n$  is  $\mathcal{E}_n$ . The second follows by induction from Theorem 1. In dimension one,  $\mathcal{E}_1$  is precisely the isotropic double exponential  $Y$ , and equality holds. Hence no smaller dimension-free base can hold simultaneously for all  $n \geq 1$ .  $\square$

## 6 Verification notes and scope

The proof is analytic and uses no numerical premise. The accompanying script `code/check_submultiplicativity.py` independently checks the explicit Gamma integral, the concavity formula for  $h'$ , the Beta splitting inequality on a finite grid, and (6) for small dimensions. These computations are smoke tests only, not part of the proof.

The principal points for human verification are:

- the scaling in (2)–(3), especially  $S_n = \sqrt{n+1} |\mathcal{E}_n| \sim \text{Gamma}(n, 1)$ ;
- the direction of the two Jensen inequalities;
- the two Beta size-bias means in Lemma 2; and
- that Theorem 4.13 of the source has exactly the full-dimensional radial log-concave scope used in Corollary 1.

Submultiplicativity does not by itself imply that  $n \mapsto A_n^{1/n}$  is non-increasing at every consecutive integer. Therefore Conjecture 2, as a stronger sequence statement, remains open in this packet. What is completely settled is the explicit question that motivates it: inequality (1), with the sharp all-dimensional base  $A_1$ .

## 7 Novelty check

Before promotion, the local run indexes were searched for arXiv:2405.07253, “one-shot separability”, “radial exponential”, and “submultiplicativity”; no overlapping result was found. A bounded web search on 21 July 2026 used the paper title, the exact one-shot phrase, “Conjecture 2”, and combinations of “Cramér transform”, “radial exponential”, and “monotonicity”. It found the arXiv and published source but no later paper resolving this question. This is a bounded search, not a claim of exhaustive novelty.

## References

- [1] S. Brazitikos and G. Chasapis, *Sharp estimates for the Cramér transform of log-concave measures and geometric applications*, Adv. Math. 478 (2025), 110407; arXiv:2405.07253.